universität innsbruck

PS 703307
Cryptography

Homework Sheet 5

Summer term 2026

In-class discussion: April 21, 2026

The instructor will first take some time to explain Assignment 2 h) of Sheet 4.

Now, we enter the dark side ...

1 Ciphertext-only attack against a substitution cipher

The following text is encrypted using a monoalphabetic substitution cipher. The plaintext is part of an English text encoded in upper case characters. Numbers and punctuation marks are left as-is. Try to recover the plaintext using the distribution of the characters in English texts.[1]

> *RB APMLFSEPULYM, U AUHKUP ARLYHP RK SBH SV FYH KROLXHKF UBI OSKF ZRIHXM NBSZB HBAPMLFRSB FHAYBRWGHK. RF RK U FMLH SV KGDKFRFGFRSB ARLYHP RB ZYRAY HUAY XHFFHP RB FYH LXURBFHJF RK PHLXUAHI DM U XHFFHP KSOH VRJHI BGODHP SV LSKRFRSBK ISZB FYH UXLYUDHF. RF RK GBNBSZB YSZ HVVHAFRTH FYH AUHKUP ARLYHP ZUK UF FYH FROH, DGF RF RK XRNHXM FS YUTH DHHB PHUKSBUDXM KHAGPH, BSF XHUKF DHAUGKH OSKF SV AUHKUP'K HBHORHK ZSGXI YUTH DHHB RXXRFHPUFH UBI SFYHPK ZSGXI YUTH UKKGOHI FYUF FYH OHKKUEHK ZHPH ZPRFFHB RB UB GBNB-SZB VSPHREB XUBEGUEH. HTHB UK XUFH UK 1915, FYH AUHKUP ARLYHP ZUK RB GKH: FYH PGKKRUB UPOM HOLXSMHI RF UK U PHLXUAHOHBF VSP OSPH ASOLXRAUFHI ARLYHPK ZYRAY YUI LPSTHI FS DH FSS IRVVRAGXF VSP FYHRP FPSSLK FS OUKFHP.*

2 Total break of a single LFSR

Consider a LFSR of degree 6 and the feedback vector $(a_0, a_1, a_2, a_3, a_4, a_5) = (1, 1, 1, 0, 0, 1)$.

a) Draw a diagram of the circuit that represents this LFSR (like on slide 16).

b) Your Assistant Cryptanalyst managed to extract the first 6 bits of the key stream using a known-plaintext attack,

$$(k_6, k_7, k_8, k_9, k_{10}, k_{11}) = (\mathtt{0}, \mathtt{0}, \mathtt{0}, \mathtt{0}, \mathtt{0}, \mathtt{1}).$$

Given this information, find the initial state $(s_0, s_1, s_2, s_3, s_4, s_5)$ by setting up a system of equations over GF(2) and solving it by hand via Gaussian elimination.

[1] https://en.wikipedia.org/wiki/Letter_frequency

3 Correlation attack against combined LFSRs

Consider three LFSRs and their feedback vectors $(a_0, a_1, \ldots, a_{\ell-1})$,

- LFSR_1 has a degree of 15 and a feedback vector of $(1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1)$,
- LFSR_2 has a degree of 16 and a feedback vector of $(1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1)$,
- LFSR_3 has a degree of 19 and a feedback vector of $(1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1)$,

and a nonlinear combiner

$$f(x_1, x_2, x_3) = x_1 \cdot x_2 \oplus x_1 \cdot x_3 \oplus x_2 \cdot x_3.$$

a) Find the period of the sequence generated by the key-stream generator. (Preferably, through analysis rather than simulation.)

b) Find the probabilities that the output of f matches the value of x_1, x_2, and x_3, respectively. It may be helpful to represent f as a truth table.

c) Implement each of the three LFSRs. To check the correctness of your implementation, compare the first 100 output bits for the seeds $(0, 0, \ldots, 0, 1)$ with the correct output given in the file `output.txt` on OLAT.

d) In OLAT, you find two key streams generated with function f and the defined LFSRs, `stream1.txt` and `stream2.txt`. (You can imagine that your assistant cryptanalyst has recovered them from a known plaintext attack.) Your task is to find the initial state of LFSR_1.